

Summary and Recommendations

- **Objective**

- The primary objective of this analysis is to examine customer churn behavior by leveraging all available features in the dataset, including demographics, account details, services subscribed, and financial attributes.

The aim is to identify patterns, correlations, and key drivers of churn, and provide data-driven recommendations to improve customer retention and business performance.

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- **Demographics Analysis:**

- **Gender** shows no significant impact on churn, indicating uniform behavior across male and female customers.
- **Senior citizens** exhibit a higher churn tendency, suggesting the need for customized retention plans for this segment.
- Customers with **dependents and partners** are less likely to churn, indicating higher stability in such groups.

- **Customer Account Information:**

- **Tenure** is a critical factor—customers with **lower tenure (new customers)** show significantly higher churn rates.
- **Contract type** plays a major role:
 - **Month-to-month contracts** → **highest churn**
 - **One-year and two-year contracts** → **lower churn**
- Customers opting for **paperless billing** show relatively higher churn behavior.

- **Payment Method Analysis:**

- Customers using **electronic check** have the highest churn rate.
- More stable payment methods (credit card, bank transfer) are associated with lower churn, suggesting better customer commitment.

- **Service Usage Patterns:**

- **PhoneService & MultipleLines:** Widely adopted, but these do not significantly differentiate churn behavior.
- **InternetService:**
 - **Fiber optic users** → **higher churn** (possibly due to cost or service expectations)
 - **DSL users** → **comparatively lower churn**
- Customers with **no internet service** show the lowest churn levels.

- **Value-Added Services (Key Insight Area):**

- Services such as:
 - OnlineSecurity
 - OnlineBackup
 - DeviceProtection
 - TechSupport
- Customers **without these services** have **significantly higher churn rates**.
- These services act as **retention drivers**, increasing customer stickiness.

- **Entertainment Services:**
 - **StreamingTV and StreamingMovies** show moderate adoption.
 - These services do not strongly reduce churn but indicate customer engagement levels.
- **Charges Analysis:**
 - Customers with **higher monthly charges** tend to churn more frequently.
 - Customers with **higher total charges (long-term users)** are more loyal and less likely to churn.
- **Overall Insight:**
 - Churn is strongly influenced by a combination of:
 - **Low tenure**
 - **Flexible contracts (month-to-month)**
 - **Lack of value-added services**
 - **Higher monthly costs**
 - Customers who are more **engaged (long tenure + multiple services)** are significantly more likely to stay.